

Problem 15.30

A loan of \$1,000 is to be repaid with annual payments at the end of each year for the next 20 years. For the first five years the payments are k per year; the second 5 years, $2k$ per year; the third 5 years, $3k$ per year; and the fourth 5 years, $4k$ per year. Find an expression for k .

Solution

Let the annual effective interest rate be i , and let

$$v = \frac{1}{1+i}.$$

The first five payments of k have present value

$$ka_{\overline{5}|}.$$

The second five payments of $2k$ are deferred 5 years, so their present value is

$$2ka_{\overline{5}|}v^5.$$

The third five payments of $3k$ are deferred 10 years, so their present value is

$$3ka_{\overline{5}|}v^{10}.$$

The fourth five payments of $4k$ are deferred 15 years, so their present value is

$$4ka_{\overline{5}|}v^{15}.$$

Therefore, the present value equation is

$$1000 = ka_{\overline{5}|} + 2ka_{\overline{5}|}v^5 + 3ka_{\overline{5}|}v^{10} + 4ka_{\overline{5}|}v^{15}.$$

Factor out $ka_{\overline{5}|}$:

$$1000 = ka_{\overline{5}|} (1 + 2v^5 + 3v^{10} + 4v^{15}).$$

Solving for k gives

$$k = \frac{1000}{a_{\overline{5}|} (1 + 2v^5 + 3v^{10} + 4v^{15})}.$$

Equivalently, since $v = (1+i)^{-1}$,

$$k = \frac{1000}{a_{\overline{5}|} (1 + 2(1+i)^{-5} + 3(1+i)^{-10} + 4(1+i)^{-15})}.$$